



HOMEWORK ASSIGNMENT

AI Assisted Software Development

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Version history:

1.0	5 October 2026	Initial version
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INTRODUCTION

Under the conditions announced for the semester, students may earn an offered grade by completing and presenting a homework assignment. The requirements are summarised below. The aim is to apply AI-assisted development practices deliberately: in addition to delivering a working application, you must demonstrate that you can verify requirements, manage changes and understand your own solution.

CHOOSING A TOPIC

You must develop an individual project. Upload a short written specification to the "Homework specification" assignment in Moodle by the end of teaching week 6, and agree on the topic with the course instructor. The submission must include 6–8 numbered, verifiable acceptance criteria, the main roles and scenarios, and any significant uncertainties and how they have been resolved. The acceptance criteria must cover normal operation, edge cases and invalid input handling. Submit the specification as text in Moodle, with a maximum length of 4,000 characters.

If you need inspiration, here are some ideas:

- Calculator (with memory, user management and user history)
- Shopping list (with user management and support for shared lists)
- Plant watering manager (user management and two-way communication between the UI and backend)
- Fitness app (user management and a more complex data model)
- Recipe app (user management, sharing and comments)
- E-book reader (user management, uploading your own e-books and bookmarks)
- Mini-game (2048, Tetris, Pac-Man, etc., with user management and a leaderboard)
- The homework assignment may also be an entry submitted to the Gránit Bank ideas competition:

<https://www.aut.bme.hu/News/PressRoom/Agentic-AI-alapu-otletpalyazat>.

Choose a small, clearly defined application. The topic must allow you to test meaningful business or operational rules and implement a small change later on; the number of screens or features is not an objective in itself.

TECHNOLOGY

You may use any technology, but the application is expected to include both a backend and a frontend. In the examples above, the backend could perform calculator operations or store the shopping list. Applications without a separate backend or frontend may also be submitted if the remaining part is correspondingly more complex (e.g. a mobile application). This must be made clear in the project specification.

DEVELOPMENT

The homework assignment must be completed independently during the semester. **Copying another person's completed assignment or sharing your own with others is prohibited.** Reused or copied code fragments will be checked using automated tools. **In cases of proven plagiarism, the**

assignments of all parties involved will be declared invalid and course sign-off or a grade will be withheld.

You must use artificial intelligence (AI) tools (such as Google Antigravity, GitHub Copilot, Claude Code, Codex, etc.) in at least three development phases, for example requirements analysis, implementation and testing. Document your use of AI in README.md or in a short document linked from it. Merely listing where AI was used is not sufficient: you must document specific decisions and how you verified them.

Manual coding and editing are permitted. You are responsible for deciding and explaining when to use AI, make manual changes or apply other verification methods. You are responsible for the correctness of the submitted code, tests and documentation, and must be able to explain them during the defence.

The assignment can be completed using the Gemini access provided to students; using Copilot is optional. A separate paid subscription or paid API is not required, but using them to complete the assignment is permitted. The application itself does not need to include an AI service.

DOCUMENTING THE USE OF AI

Briefly document three substantive decision cases. For each case, identify the problem, the tool, the relevant context and key instruction given to the AI, the essence of its suggestion, your decision and the reasons for it. Refer to the code, test or execution result that provides evidence of verification. The same evidence may support more than one related requirement.

Implement at least one of the requirements twice using two separate AI tools (e.g. Antigravity and Autopilot). Compare and document the results.

TESTING REQUIREMENTS

Assign a verification method to each acceptance criterion. Use automated tests to check important rules implemented in code; for purely visual or manual checks, provide reproducible steps and the expected result. The assessment guide lists the required cases.

For at least one test, retain both the faulty and corrected versions, together with the results of both test runs. If you have not found a suitable defect in your own work, you may use a clearly labelled version with a deliberately introduced fault, or request a short faulty code fragment from the instructor. Do not present this as a defect that actually occurred during development. A separate application is not required for this case.

SUBMISSION

MILESTONES AND DEADLINES

There are three submission milestones. Exact dates, submission locations and resubmission arrangements will be published in Moodle.

1. Specification: submit the topic description and 6–8 acceptance criteria, clarify significant uncertainties and obtain approval for the topic. **Deadline: end of week 6.**

2. First working version: submit the source code implementing the main user workflow, setup instructions and basic tests. Document the project state and development history in a public GitHub repository. **Deadline: end of week 10.**

3. Final version: submit the source code implementing the change request, the updated specification, tests and results, setup instructions, decision cases and a short summary of changes, and present the results. **Deadline: week 13.**

CHANGE REQUEST ISSUED AFTER THE FIRST VERSION

After the first working version has been submitted, the instructor will issue a small change request relevant to the approved topic. The change will affect an existing rule, workflow or data-handling process; it will not require a complete rewrite or a change of technology. At least two weeks will be available before the final submission deadline.

Implement the change starting from the recorded first version. Show which requirements and code sections it affected, why you chose your solution and which regression checks demonstrate that existing behaviour has been preserved. If the application uses persistent data, also verify that the data is preserved.

Example: in an equipment booking application, two overlapping bookings for the same item must not be allowed, but consecutive bookings with no gap between them are permitted; previously stored data must be preserved.

FINAL SUBMISSION AND INDIVIDUAL DEFENCE

Upload the completed assignment's source code and the materials listed below to the course's Moodle page, then demonstrate the working application at an agreed time at the end of the semester. A laboratory computer can be provided on request in advance, but using your own computer (laptop, remote desktop, etc.) will help the demonstration run smoothly. During the demonstration, you must present the following to the assessor:

- The specification and numbered acceptance criteria
- The working application
- The first and final versions of the application's source code, and a short summary of the changes
- Setup and test execution instructions, the tests and the verification results
- A general description of AI use (chosen tool, model and working method), together with three documented decision cases
- Short evaluation of alternative AI tools that were used to implement a feature twice.

The assessor will award points on the spot. During the 8–10-minute individual defence, you will present a selected requirement and its verification (approximately 2 minutes), explain part of your own code and predict its behaviour (approximately 3 minutes), and then, if requested by the assessor, perform a small modification or debugging task (approximately 3–5 minutes). AI may not be used during the explanation, but is permitted during the modification task. The task will relate to the submitted solution and the published requirements. A correct diagnosis and verification plan will earn partial credit even if the modification is not completed within the time limit. An AI service outage or slow response will not, by itself, result in a deduction; in that case, explain your modification and verification plan.

Students who wish to improve their previous score through subsequent changes to their application may also take part in the resubmission session. The improved elements and their associated checks must be demonstrated again; the defence may also be repeated. Arrangements and deadlines for making up missed milestones are provided in Moodle.

ASSESSMENT SCHEME

Assessors will evaluate submitted assignments using the following scheme.

Criterion	Maximum points
Specification and acceptance criteria	10
Working application	25
Testing and defect detection	15
AI use and decisions	20
Implementing the change request	15
Individual defence	15
Total	100

Assessors award points separately for each subcriterion below; their sum gives the category score. For each subcriterion, missing or substantially incorrect work earns 0 points, partial fulfilment earns partial credit, and complete fulfilment supported by demonstrated evidence earns the maximum. Page count, the number of prompts and the AI subscription used are not assessment criteria. The semester's course requirements specify how points are converted to grades and the other conditions for earning an offered grade.

SPECIFICATION AND ACCEPTANCE CRITERIA

The specification may be prepared in any format; a Markdown (md) file is a practical choice. There is no required page count; the content must be concise, clear and verifiable, with a maximum length of 4,000 characters.

- 3 points – The objective, roles and main scenarios are clear.
- 5 points – The 6–8 numbered acceptance criteria associate a specific input or situation with an observable expected result, covering normal operation, edge cases and error cases.
- 2 points – Significant uncertainties and the chosen interpretation are documented, and the later requirement change is incorporated.

WORKING APPLICATION

- 10 points – The agreed core features work end to end and meet the acceptance criteria.
- 5 points – Important invalid inputs and edge cases are handled correctly.
- 5 points – The application's components work together; data or state management behaves as specified.
- 5 points – The solution is clearly structured, the interface is usable and no defect prevents completion of the main user workflow.
- The maximum for the working application category is 25 points. Assessment is based on the architecture approved in advance, including applications without a separate backend and frontend. Simply being able to launch the application does not earn full marks.

TESTING AND DEFECT DETECTION

- 5 points – Checks are mapped to acceptance criteria, and executable automated tests cover important rules.
- 5 points – At least one normal case, two meaningful edge cases, one check of invalid input or an unauthorised operation, and one regression case are demonstrated.
- 5 points – Execution results demonstrate that at least one test fails on a known faulty version and passes on the corrected version. The defect and expected behaviour are explained.
- The maximum for testing and defect detection is 15 points. The number of tests, successful execution or a coverage percentage alone is not sufficient for full marks.

AI USE AND DECISIONS

The maximum for AI use and decisions is 20 points. The three decision cases are assessed using the associated code, tests or execution results.

- 3 points – AI use is demonstrated in at least three development phases; the tools used and the relevant context are identifiable.
- 3 points – The decision cases justify accepting, modifying or rejecting AI suggestions on specific technical grounds.
- 4 points – The correctness of the decisions is supported by specific checks, and the limitations of the method are explained. On the MSc course, 2 of these points are awarded for comparing two working methods and 2 for verifying the decision cases.
- 5 points – The codebase contains elements that support AI-assisted development (e.g. AGENTS.md and, where applicable, definitions of custom agents, skills, commands, etc.), and these are used correctly.
- 5 points – One of the features was implemented twice with two separate AI tools and the student has correctly evaluated the differences in results.
- You do not have to find an AI error or present a rejected suggestion: meaningful verification of a correct suggestion is equally valid. A complete conversation history, a large number of prompts or a separate paid tool is not required, but is permitted.

IMPLEMENTING THE CHANGE REQUEST

- 6 points – The new requirement is met, and the modification can be traced from the recorded first version.
- 3 points – The affected requirements and code sections are identified, and the solution is justified.
- 6 points – Regression checks demonstrate that existing behaviour and, where relevant, stored data have been preserved.

INDIVIDUAL DEFENCE

- 10 points – The behaviour of your own code and the expected result for a selected input are explained correctly.
- 5 points – An appropriate test or debugging method is selected, and its result is interpreted.